

Arizona to Utah: A Neural Network Analysis of Former Coyotes Player Performance After Relocation

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Project: Single Layer Neural Network

1. Introduction

When I mention that I love hockey, one of the first questions that people ask me is “How did you feel when the Coyotes were relocated?” The question stung at first, as it was hard for me to believe that anyone would be happy about their relocation. However, now that time has passed, and the emotional wound has begun to heal, I realize why their relocation was necessary. For instance, the Utah Mammoth made the playoffs this season, and I cannot imagine if they were still in Arizona, having playoff games in Mullett Arena in front of 5,000 fans. When the Mammoth made the playoffs, I saw that many Coyotes fans were understandably upset and felt that their team was taken right when they started improving. I even saw discussions about how this was still Arizona’s team, as many of the players are the same. However, many Utah fans argue that the team never would have succeeded in Arizona and that the move to Utah was necessary for team improvement.

Thus, I wondered, did Arizona hold these players back, or was their success in Utah due to natural development? It was an interesting question, especially since I felt like it was something that could be answered through modeling. I used to write about topics like this when I was younger, but just by using basic stats. I was very excited to revisit hockey statistics now, when I have the tools to model expected performance and actually test it. Although the results of this analysis will not change the past, or the future, I hope that

maybe it can bring some closure to Coyotes fans. If I find that players performed noticeably better in Utah while accounting for natural development, Coyotes fans could be at peace knowing the Utah move was the best thing for the players. If I find that the alternative is true, then we'll know that Arizona itself was not limiting the players, and maybe that hockey can work here.

To evaluate these two options, I started by collecting season-level statistics for the 11 players who had at least one 50+ game season with each franchise. To separate genuine performance differences from improvements due to getting older or more ice time, I decided to build a neural network that established expected performance baselines and compared residuals between Arizona and Utah.

2. The Model

The neural network model is feedforward, as it has the inputs, one hidden layer and then the output. The inputs, age, years in system (how many years since they joined the organization), average ice time, and prior PPG (points per game) goes through a hidden layer to predict point per game for that season. In this hidden layer, there are $K=10$ hidden units which take the inputs, combine them with the learned weights and pass the results through an activation function. Tanh, the activation function used, transforms each value into a number between -1 and 1 so that the network can capture curved, nonlinear relationships and it tends to work better than sigmoid for gradient computations (McCulloch, 2025). Each hidden unit captures a pattern in the data that it finds during training. The inputs contribute differently to each hidden unit, as not every input matters equally to every unit. Then, the outputs of all 10 hidden units are combined and multiplied by a learned weight to sum into a final prediction, in this case, expected PPG. Each layer of the network applies a nonlinear transformation to the outputs of the previous layer, and the full network represents the composition of all these transformations (McCulloch, 2025).

Specifically, I chose a neural network model because it can express curved relationships. A linear regression assumes that the relationship between the inputs and output is a straight line, which is unrealistic in hockey where development is rarely linear. Players often improve or

decline at complex, nonlinear rates. So, by using a neural network, I hope to draw conclusions that represent the situation more accurately than surface-level observations like a player scoring more on one team, or the team winning more.

3. The Algorithm/Method

Loss Function

The methodologies described in this section are based on the concepts in section 10.7 of *Introduction to Statistical Learning* by Gareth James, et al. For a neural network, mini-batch gradient updates are used to train the model to become more accurate. Each prediction is evaluated using MSE, mean squared error, as each output is subtracted by the real PPG and the difference is squared and averaged across all predictions. The network is continuously updating to minimize MSE.

Stochastic Gradient Descent

This is part of the process of Stochastic Gradient Descent. The gradient is the mathematical direction of steepest increase in MSE, so the network moves the weights in the opposite direction to reduce it. This process repeats for a set number of epochs, in my case 500, until the loss stabilizes. It uses random mini-batches instead of all the data at once as it is faster.

Regularization

L2 regularization was also applied, adding a penalty for large weights during training to prevent the network from overfitting on the small dataset.

Simulated Data Validation

Before applying this model to hockey data, I validated the neural network implementation on simulated data to test the effect of varying the number of hidden units, activation function, learning rate, and regularization parameter.

Hyperparameter Selection

To be more specific on the hyperparameter choices I used, I started with $K=10$ hidden units because it is enough for the network to capture patterns, but it avoids overfitting on my small dataset. Since I only have 11 players with a limited number of seasons, a smaller K is safer. I tested different K values ($K=2,5,20$) for the simulated data and saw that more units captures more complexity, but 10 is reasonable to avoid overfitting. For the learning rate, I tested values of 0.001, 0.01, and 0.1. While 0.1 performed well on the simulated data, faster learning rates can become unstable on small noisy datasets. A learning rate of 0.01 was chosen as a more conservative choice for the hockey data. Similarly, for regularization, I tested lambda values of 0.0001, 0.01, and 0.5. While the lower lambda fit the simulated data best, my hockey dataset is small with only 11 players and limited seasons. A higher lambda of 0.01 was chosen to reduce the risk of the network overfitting to this specific group of players rather than learning patterns.

Data Preparation

Season-level statistics were collected from Hockey Reference for 11 former Arizona Coyotes players who had at least one 50+ game season with both franchises. The 50+ game threshold was chosen because anything less than half a season is likely affected by injury or trades, making the statistics unreliable for comparison.

The variables included were PPG, prior season PPG, age, average time on ice, and years in system. Points per game captures production while accounting for varying games played. Prior PPG was included because players tend to perform at similar levels year over year. Age was included because younger players naturally develop over time while older players may decline, regardless of team. Average time on ice matters because more ice time directly affects scoring opportunity. Years in system was included to capture whether familiarity or stagnation within an organization affects performance. All input features were scaled before training so that variables with larger numeric ranges, like ice time, would not dominate over smaller ones like PPG. The data was then split 80% for training and 20% for testing, which is necessary because evaluating a model on the same data it was trained on overstates its

true predictive performance.

4. Results

Simulated Data Results

I started by using the simulated data to test how different settings affect how well the neural network learns. I used a sine wave as the original data because it would be clear if the model matched the pattern or not. The sine wave plot that used linear regression was clearly inadequate as a straight line cannot represent a curve. This demonstrates the need to use a neural network.

When creating the model, I tried changing the amount of units. For 2 units, the network is too simple and cannot fit the curve accurately. The other graphs for 5 units and 20 units both resemble the curve. They are so close that there is no noticeable differences in the two plots, but more units do allow for greater flexibility to capture the complex patterns. As for which activation function to use, I chose tanh because it fits the curve more cleanly with the same number of units.

For the training stability, I found that for regularization, a very low lambda (0.0001) fit the simulated data the best, since lambda increasing led to the model underfitting the data. For the learning rate, I tested 0.001, 0.01, and 0.1 and noticed that 0.1, the fastest, converges the best in 1000 epochs. However, for the hockey data a slightly higher lambda and a slightly lower learning rate were chosen given the small dataset size.

Hockey Data: Descriptive Results

The scatterplot of PPG before and after relocation shows a mixed picture. Players like Clayton Keller and Logan Cooley scored notably higher in Utah, while players like Matias Maccelli and Alexander Kerfoot had higher PPG in Arizona. Several players, including Barrett Hayton and Jack McBain, performed at roughly the same level in both locations. The percent change chart reinforces this as there is no clear directional trend, with some players showing large increases, others showing decreases, and several staying relatively flat.

However, these are raw scoring numbers that do not account for age or natural development. A younger player scoring more in Utah may simply be improving due to experience rather than due to the relocation. This is why a neural network model is needed as it establishes expected performance baselines to isolate whether players in Utah produced beyond what development alone would predict.

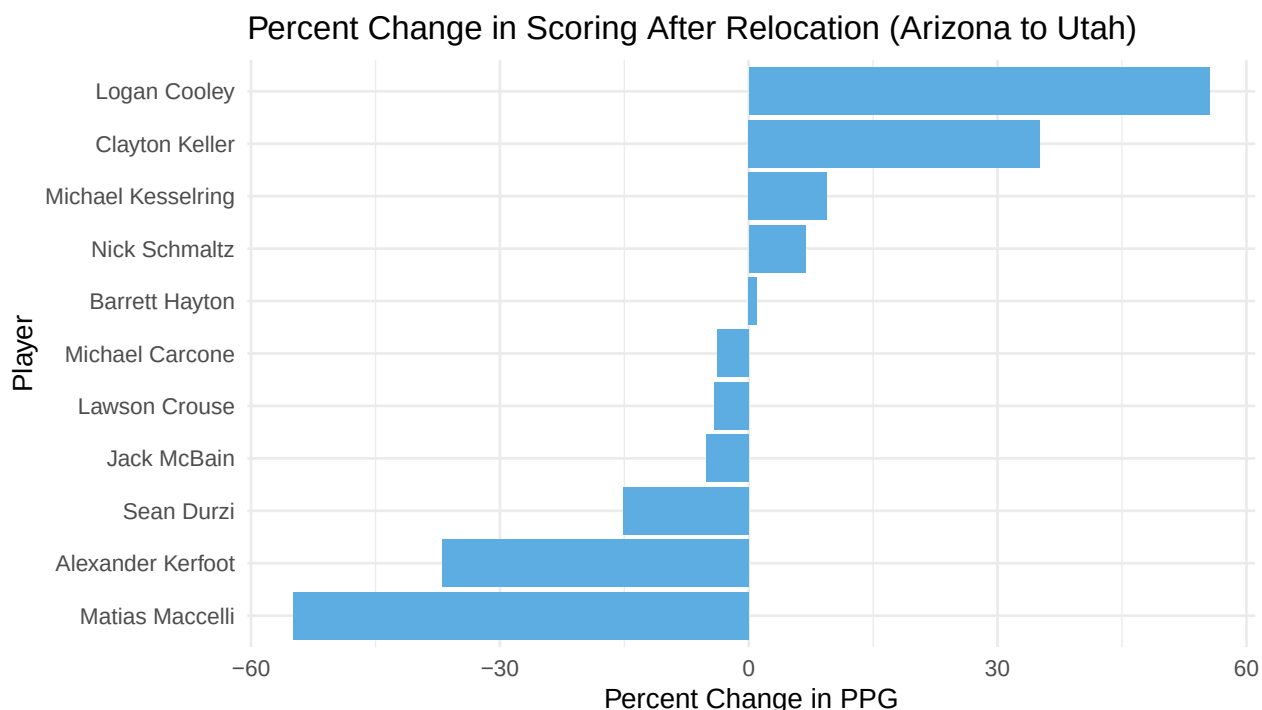


Figure 1: Percent Change in Scoring After Relocation

Hockey Data: Neural Network Results

The neural network was evaluated using R^2 and MSE on both training and test data. R^2 (R-squared) measures how much of the variation in PPG the model explains. The model achieved a training R^2 of 0.771, meaning the four inputs explain 77% of the variation in PPG. On the held-out test data this dropped to 0.560, suggesting the model learned some patterns that are specific to the training data rather than fully general. Despite this overfitting, the model still has meaningful predictive power.

The MSE (Mean Squared Error) is the average squared prediction error. Train MSE=0.014 and Test MSE=0.017 are very close to each other, which means the model is not significantly

less accurate than the training set. Since PPG values are small numbers (typically between 0 and 1.2), an error of 0.014-0.017 means the predictions are off by about 0.12-0.13 PPG on average which is reasonable. The training loss curve confirmed the model was learning, with MSE decreasing steadily across all 500 epochs.

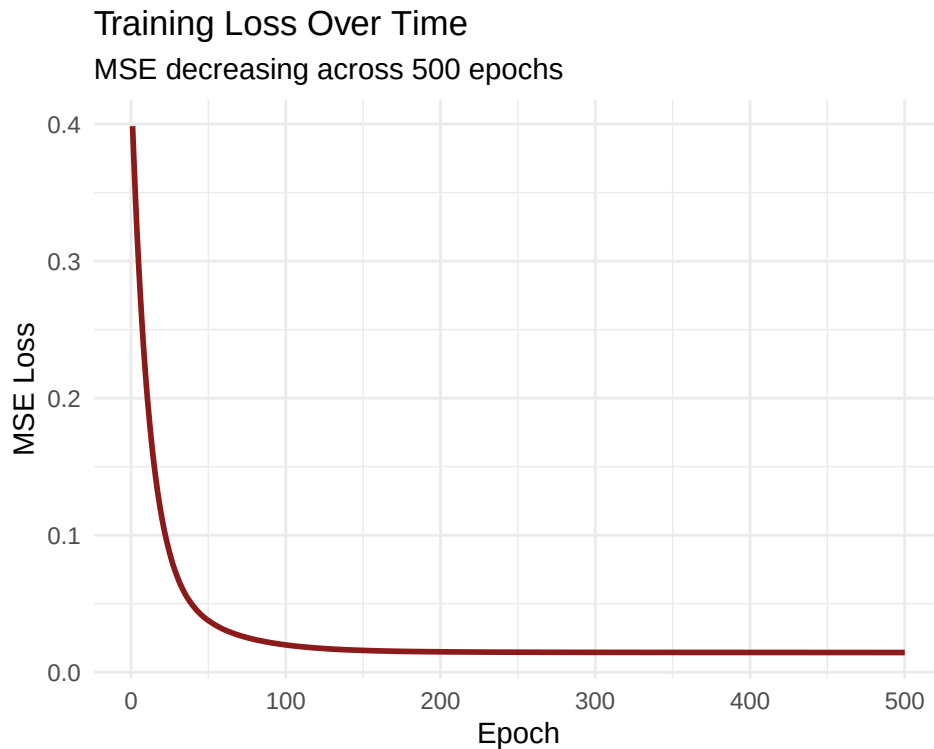


Figure 2: Training Loss Over Time — MSE decreasing across 500 epochs

The actual vs predicted plot shows no clear clustering of Arizona or Utah seasons above or below the line, with most player seasons falling close to model predictions regardless of team.

The residuals boxplot reinforces this as Arizona seasons show a slightly positive median residual, meaning players marginally outperformed expectations, while Utah seasons show a slightly negative median. However, both distributions are centered close to zero with overlapping ranges. Utah shows greater variation, likely reflecting the limited data of only two seasons. Overall the model does not reveal a strong systematic difference in performance relative to expectations between the two franchises.

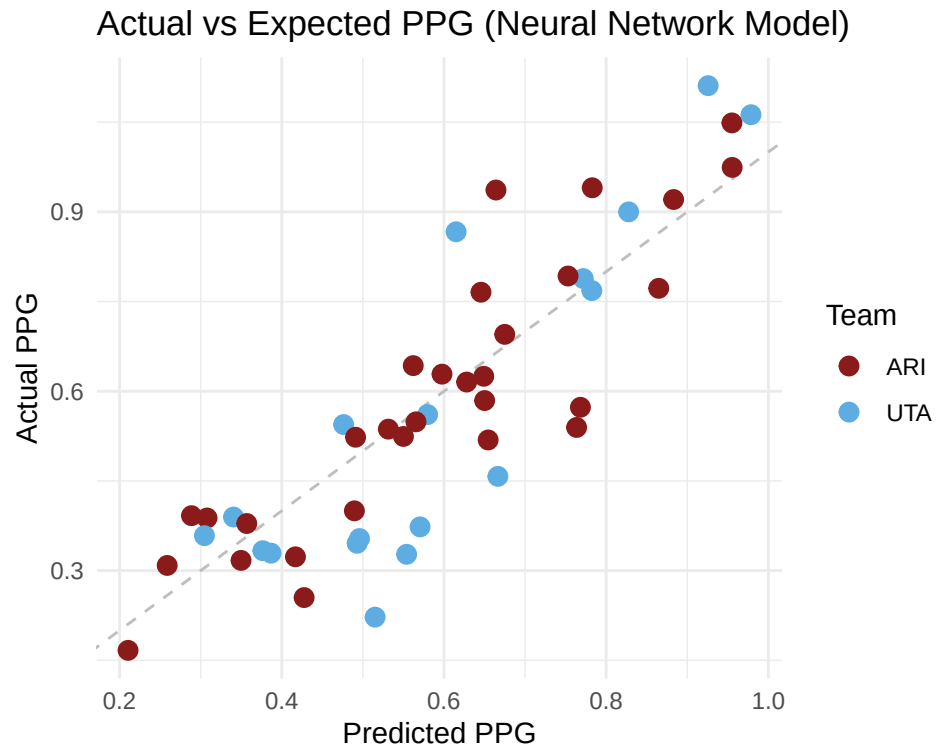


Figure 3: Actual vs Expected PPG - Neural Network Model

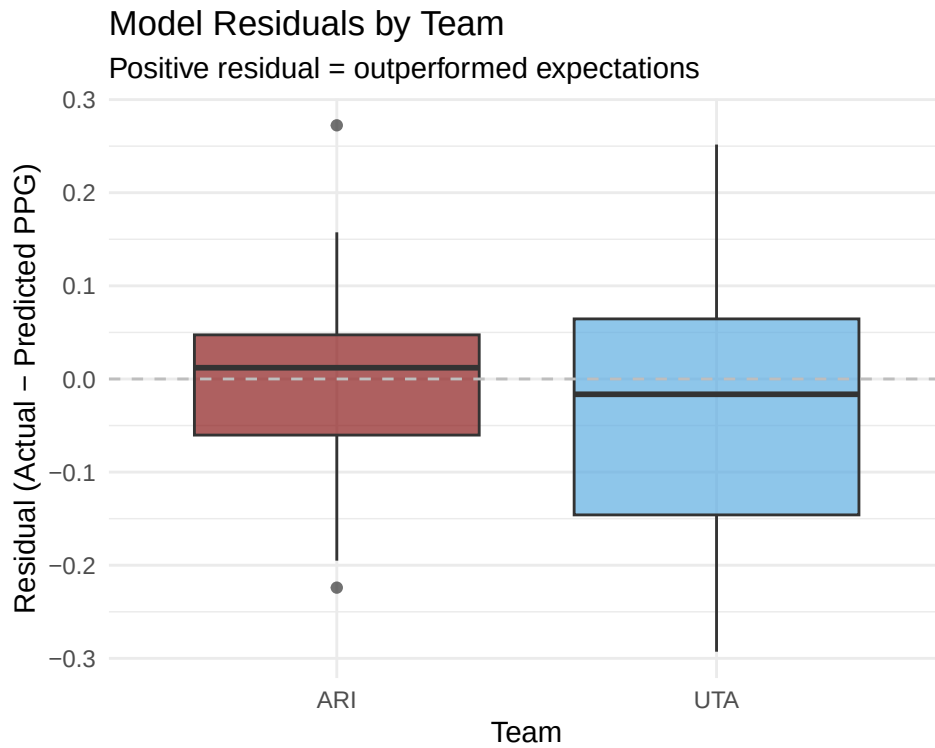


Figure 4: Model Residuals by Team - Positive residual indicates outperforming expectations

5. Discussion

Based on the analysis, the model does not support the conclusion that Arizona held these players back. The residuals show no meaningful difference in performance between the two franchises. There was no clear clustering of Arizona or Utah data points above or below the predicted PPG line. Although certain players produced more than predicted based on their prior PPG, age, average ice time, and years in system, it was not a majority of the players in the analysis. Thus, there is not enough evidence that the relocation to Utah improved player performance.

While the above finding is nothing monumental, I am pleasantly surprised by the smaller stories that came from this analysis. If I examine key players, such as Keller, I see that he has consistently outperformed expectations in both places but especially in Utah. I am surprised by Cooley, as I thought he would exceed predictions in Utah, and he did in one season, but in the other he performed as expected. It is difficult to draw conclusions when looking at such young players. I can see clearly here what is also obvious from Crouse's career arc. He has been very consistent in Arizona, performing as expected, but severely declined his first season in Utah and now has bounced back. This was possibly an adjustment period for him, as he adapted to a new city and organization.

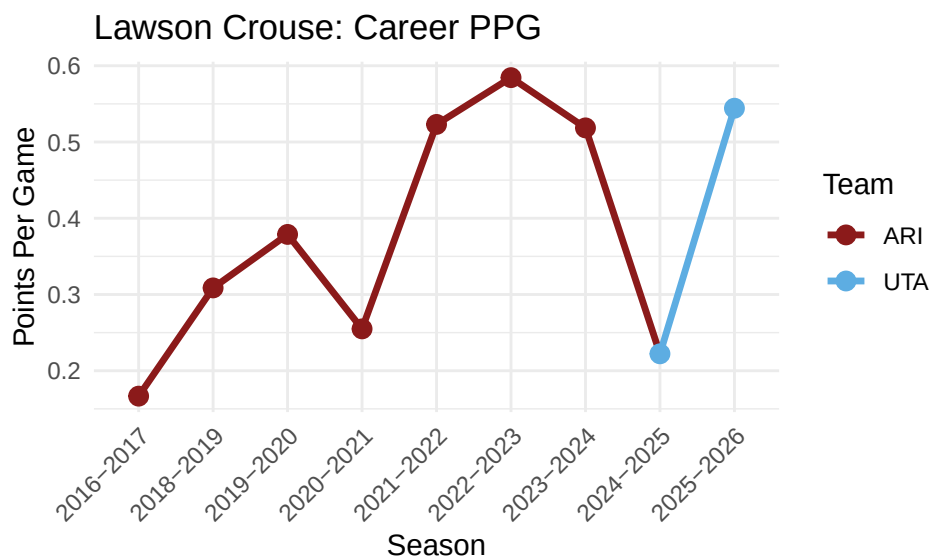


Figure 5: Lawson Crouse Career PPG - Arizona vs Utah

Another thing I learned from this analysis is that there are severe limitations worth noting. The dataset is imbalanced because Utah only has 2 seasons of data which explains the larger variation in the residual plot. On the contrary, in Arizona there were several years where the team was extremely bad which influenced several players' production. Also, players like Cooley complicate the findings as they are young and their production can vary due to many factors. Only 56% of the variation is explained by my model, so a large portion of the results are affected by other variables such as coaching, line combinations, personal factors, etc. Thus, it is difficult to draw large conclusions from the model, but the small ones I can make are nonetheless fascinating.

For me, it was healing to return to sports analytics for this project because I feel closure knowing that the move to Utah has not yet proven to be transformative. It was not Arizona itself that led to poor production for many players over the last decade. I hope this work could encourage Coyotes fans to believe that hockey can work in Arizona, and that the team in Utah is successful largely due to the work that was done here, in the valley of the sun.

6. References

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning: With Applications in R*. Springer.

McCulloch, R. (2025). Neural Networks. Arizona State University. https://www.rob-mcculloch.org/2025_cs/webpage/notes/nnet.pdf

Hockey Reference. (2025). *Player Statistics*. Retrieved April 2026, from <https://www.hockey-reference.com>

7. Appendix

Full R code, simulated data experiments, and all plots supporting this analysis are included in the appendix. All code was written in R using the tidyverse and ggrepel packages